

K L UNIVERSITY
FRESHMAN ENGINEERING DEPARTMENT
A Project Based Lab Report
On
Smart health & Medical History Tracker + Wellness Analytics
System

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CERTIFICATE

This is to certify that the project based laboratory report entitled “**Smart health & Medical History Tracker + Wellness Analytics System**” submitted Farheen Fathima, Ch. Deva Harsha, V.Ashmitha, N. Krishna Mohan bearing Regd. No-2300031597, 2500031422 , 2500031670, 2500031693 to the **DEPARTMENT of BASIC ENGINEERING SCIENCE-I, K L UNIVERSITY** in partial fulfillment of the requirements for the completion of a project based Laboratory in “**PYTHON PROGRAMMING (25SC1204)**” course in I B Tech II Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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HEAD OF THE DEPARTMENT

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ABSTRACT

The **Smart Health & Medical History Tracker + Wellness Analytics System** is a Python-based application developed to provide an efficient and user-friendly solution for managing personal health records digitally. In modern healthcare, maintaining accurate and accessible medical history is essential for timely diagnosis and treatment. However, traditional methods of record-keeping are often unorganized, prone to loss, and difficult to access when needed. This project addresses these challenges by offering a centralized platform for storing and analyzing health-related data.

The system allows users to input, update, and retrieve important medical information such as personal details, past illnesses, medications, allergies, and daily health parameters like temperature, weight, and blood pressure. By organizing this data systematically, the application ensures easy access and better management of health records.

A key feature of this project is its **wellness analytics capability**, which enables users to analyze their health data over time. The system can identify trends, generate summaries, and provide basic insights that help users monitor their health conditions and make informed decisions. This promotes proactive healthcare and increases awareness about personal well-being.

The application is implemented using Python, incorporating fundamental programming concepts such as file handling, data structures, and modular design. These features ensure that the system is efficient, scalable, and easy to maintain.

In conclusion, this project demonstrates how a simple yet powerful digital solution can enhance personal health management by combining data storage with analytical insights. It serves as a foundation for future improvements such as cloud integration, mobile applications, and real-time health monitoring systems.

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OBJECTIVES

The objectives of the Smart Health & Medical History Tracker + Wellness Analytics System are designed to ensure effective health data management and meaningful analysis. The key objectives of this project are as follows:

- To develop a digital platform for storing and managing personal health and medical history in an organized manner.
- To maintain accurate medical records including past illnesses, treatments, medications, allergies, and doctor details.
- To enable easy data entry and retrieval, allowing users to quickly access their health information whenever required.
- To track daily health parameters such as body temperature, weight, blood pressure, and other vital signs.
- To analyze health data and identify patterns or trends over time for better health awareness.
- To provide basic wellness analytics, helping users understand improvements or declines in their health condition.
- To generate simple reports or summaries that can assist both users and healthcare professionals.
- To minimize manual errors and data loss by replacing traditional paper-based records with a digital system.
- To implement core Python concepts such as file handling, functions, and data structures in a real-world application.
- To design a user-friendly interface that can be easily used by individuals with minimal technical knowledge.
- To promote preventive healthcare, encouraging users to regularly monitor and manage their health.

INTRODUCTION

The **Smart Health & Medical History Tracker + Wellness Analytics System** is a Python-based application designed to help individuals efficiently manage their personal health data and medical history in a structured and intelligent manner. In today's fast-paced world, maintaining accurate health records is often neglected, leading to difficulties in tracking past illnesses, medications, and overall wellness trends. This system addresses that problem by providing a centralized digital platform for storing, updating, and analyzing health-related information.

The application allows users to record essential medical details such as personal information, past medical history, ongoing treatments, allergies, prescriptions, and daily health metrics like weight, temperature, and blood pressure. Unlike traditional record-keeping methods, this system ensures that data is organized, easily accessible, and can be updated in real time.

In addition to basic data storage, the system incorporates **wellness analytics**, which helps users gain insights into their health patterns over time. By analyzing stored data, the system can identify trends, generate summaries, and assist users in making informed health decisions. For example, users can observe fluctuations in their health parameters and take preventive measures when necessary.

The project is developed using Python, utilizing concepts such as file handling, data structures, and modular programming. It may also include graphical representation of health data for better visualization and understanding. The simplicity and flexibility of Python make it an ideal choice for implementing such a system.

AIM

The primary aim of the Smart Health & Medical History Tracker + Wellness Analytics System is to develop an efficient and user-friendly application that helps individuals manage, store, and analyze their health-related data in a systematic and secure manner.

The specific objectives of this project include:

- To create a centralized platform for storing personal medical history, including illnesses, treatments, medications, and allergies.
- To enable users to easily add, update, and retrieve their health records whenever required.
- To track daily health parameters such as weight, temperature, and other vital indicators.
- To provide basic **data analytics** features that help users understand their health trends over time.
- To generate simple reports and summaries that can assist users and doctors in decision-making.
- To reduce dependency on manual record-keeping and minimize the risk of losing important medical information.
- To implement the system using Python programming concepts such as file handling, functions, and data structures.
- To ensure the system is simple, efficient, and accessible for users with basic technical knowledge.

Through this project, the aim is not only to store health data but also to transform it into meaningful insights that promote better health awareness and proactive care.

ADVANTAGES

- Easy to use and understand
- Stores patient health data efficiently
- Provides quick health suggestions
- Helps in maintaining medical records
- Saves time compared to manual tracking

DISADVANTAGES

- Stores data only temporarily (no database)
- Limited health analysis
- No graphical user interface (GUI)
- Not suitable for large-scale hospital use
- Requires manual data entry

SYSTEM REQUIREMENTS:

Software Requirements

- Python (3.x version)
- Any IDE (IDLE / VS Code / PyCharm)
- Operating System: Windows / Linux / Mac

Hardware Requirements

- Computer or Laptop
- Minimum 2GB RAM
- Keyboard for input
- Monitor for display

IMPLEMENTATION

```
import json
import os
import matplotlib.pyplot as plt

FILE_NAME = "patients.json"

# ----- FILE -----
def load_data():
    if os.path.exists(FILE_NAME):
        with open(FILE_NAME, "r") as f:
            try:
                return json.load(f)
            except:
                return {}
    return {}

def save_data(data):
    with open(FILE_NAME, "w") as f:
        json.dump(data, f, indent=4)

# ----- INPUT -----
def get_input(msg):
    val = input(msg)
    while val.strip() == "":
        print("Cannot be empty!")
        val = input(msg)
    return val

def get_float(msg):
    while True:
        try:
            return float(input(msg))
        except:
            print("Enter number!")

# ----- BMI -----
def calculate_bmi(w, h):
    return round(w / (h*h), 2)

def bmi_status(b):
```

```

    if b < 18.5: return "Underweight"
    elif b < 25: return "Normal"
    elif b < 30: return "Overweight"
    else: return "Obese"

# ----- ADD PATIENT -----
def add_patient(data):
    print("\n--- ADD PATIENT ---")
    pid = get_input("ID: ")
    if pid in data:
        print("Already exists!"); return

    name = get_input("Name: ")
    age = get_input("Age: ")
    gender = get_input("Gender: ")

    data[pid] = {
        "name": name,
        "age": age,
        "gender": gender,
        "records": []
    }
    print("Patient Added!")

# ----- ADD RECORD -----
def add_record(data):
    print("\n--- ADD RECORD ---")
    pid = get_input("ID: ")

    if pid not in data:
        print("Not found!"); return

    date = get_input("Date: ")
    weight = get_float("Weight: ")
    height = get_float("Height: ")
    bp = get_input("BP: ")
    sugar = get_input("Sugar: ")
    notes = get_input("Notes: ")

    bmi = calculate_bmi(weight, height)
    status = bmi_status(bmi)

    record = {

```

```

        "date": date,
        "weight": weight,
        "height": height,
        "bmi": bmi,
        "status": status,
        "bp": bp,
        "sugar": sugar,
        "notes": notes
    }

    data[pid]["records"].append(record)
    print("Record Added!")

# ----- VIEW -----
def view_all(data):
    print("\n--- ALL PATIENTS ---")
    if not data:
        print("Empty"); return
    for pid in data:
        print(pid, "-", data[pid]["name"])

def view_one(data):
    print("\n--- VIEW PATIENT ---")
    pid = get_input("ID: ")

    if pid not in data:
        print("Not found!"); return

    p = data[pid]
    print("\nName:", p["name"], "Age:", p["age"], "Gender:", p["gender"])

    if not p["records"]:
        print("No records"); return

    for r in p["records"]:
        print("\nDate:", r["date"])
        print("BMI:", r["bmi"], r["status"])
        print("BP:", r["bp"], "Sugar:", r["sugar"])
        print("Notes:", r["notes"])

# ----- GRAPH FUNCTION -----
def show_graph(data):
    print("\n--- GRAPH ANALYSIS ---")

```

```

pid = get_input("Enter Patient ID: ")

if pid not in data:
    print("Patient not found!")
    return

records = data[pid]["records"]

if len(records) < 2:
    print("Need at least 2 records for graph!")
    return

dates = []
weights = []
bmis = []

for r in records:
    dates.append(r["date"])
    weights.append(r["weight"])
    bmis.append(r["bmi"])

# Weight Graph
plt.figure()
plt.plot(dates, weights)
plt.title("Weight Trend")
plt.xlabel("Date")
plt.ylabel("Weight")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# BMI Graph
plt.figure()
plt.plot(dates, bmis)
plt.title("BMI Trend")
plt.xlabel("Date")
plt.ylabel("BMI")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# ----- ANALYSIS -----
def analysis(data):

```

```

print("\n--- ANALYSIS ---")
pid = get_input("ID: ")

if pid not in data:
    print("Not found!"); return

recs = data[pid]["records"]

if not recs:
    print("No records"); return

last = recs[-1]

print("\nLatest BMI:", last["bmi"])
print("Status:", last["status"])

# ----- DELETE -----
def delete(data):
    print("\n--- DELETE ---")
    pid = get_input("ID: ")
    if pid in data:
        del data[pid]
        print("Deleted")
    else:
        print("Not found")

# ----- MENU -----
def menu():
    print("\n===== MENU =====")
    print("1.Add Patient")
    print("2.Add Record")
    print("3.View All")
    print("4.View One")
    print("5.Graphs")
    print("6.Analysis")
    print("7.Delete")
    print("8.Save & Exit")

# ----- MAIN -----
def main():
    data = load_data()

    while True:

```

```
menu()
ch = input("Choice: ")

if ch == "1": add_patient(data)
elif ch == "2": add_record(data)
elif ch == "3": view_all(data)
elif ch == "4": view_one(data)
elif ch == "5": show_graph(data)
elif ch == "6": analysis(data)
elif ch == "7": delete(data)
elif ch == "8":
    save_data(data)
    print("Saved!")
    break
else:
    print("Invalid!")

if __name__ == "__main__":
    main()
```

INTEGRATION ANDVSYSYSTEM TESTING OUTPUTS

Screenshot:

```
"C:\Users\Farheen Fathima\PyCharmMiscProject\.venv\Scripts\python.exe" "C:\Users\Farheen Fathima\PyCharmMiscProject\main.py"

===== MENU =====
1.Add Patient
2.Add Record
3.View All
4.View One
5.Graphs
6.Analysis
7.Delete
8.Save & Exit
Choice: |
```

```
8.Save & Exit
Choice: 1

--- ADD PATIENT ---
ID: 123
Name: sreeja
Age: 23
Gender: female
Patient Added!

===== MENU =====
1.Add Patient
2.Add Record
3.View All
4.View One
5.Graphs
6.Analysis
7.Delete
8.Save & Exit
Choice: 2

--- ADD RECORD ---
ID: 123
Date: 12-3-2026
Weight: 45
Height: 5
BP: no
Sugar: no
Notes: has some allergy
```



```
--- ADD RECORD ---
```

```
ID: 123
```

```
Date: 12-3-2026
```

```
Weight: 45
```

```
Height: 6
```

```
BP: no
```

```
Sugar: yes
```

```
Notes: noo
```

```
Record Added!
```

```
===== MENU =====
```

```
1.Add Patient
```

```
2.Add Record
```

```
3.View All
```

```
4.View One
```

```
5.Graphs
```

```
6.Analysis
```

```
7.Delete
```

```
8.Save & Exit
```

```
Choice: 3
```

```
--- ALL PATIENTS ---
```

```
123 - sreeja
```

```
===== MENU =====
```

```
1.Add Patient
```

```
2.Add Record
```

```
3.View All
```

CONCLUSION

The **Smart Health & Medical History Tracker + Wellness Analytics System** successfully demonstrates how technology can be utilized to improve personal health management. By integrating data storage with analytical features, the system provides a comprehensive solution for maintaining and understanding medical records.

This project highlights the importance of digital healthcare solutions in modern life. It simplifies the process of recording and accessing medical history, ensuring that important health information is always available when needed. The inclusion of wellness analytics adds significant value by allowing users to monitor trends and make informed decisions about their health.

The use of Python programming in this project showcases its effectiveness in building practical and real-world applications. Concepts such as file handling, modular design, and data processing are effectively applied to create a functional and efficient system.

Although the current system provides essential features, it can be further enhanced in the future by adding advanced functionalities such as cloud storage, mobile application integration, real-time health monitoring through IoT devices, and improved data visualization techniques.

In conclusion, this project serves as a step towards smarter healthcare management by combining simplicity, functionality, and analytical capabilities. It not only helps users maintain their medical records but also encourages a more proactive and informed approach to personal wellness.